
MoSAIC: Modular Self-Attentive Interacting Columns for Recurrent Memory

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Abstract

Recurrent models preserve a fixed-size state, but conventionally organize it as one monolithic stream per layer. MoSAIC (Modular Self-Attentive Interacting Columns) instead factorizes state into persistent GRU columns and routes information among them with attention. We study this organization on Store-Distract-Query (SDQ) and text8 under a one-billion-token training target, re-aggregating historical results at common logged budgets. At approximately ten million parameters, L2C4 obtains 0.839 ± 0.006 SDQ Acc++ versus 0.541 ± 0.073 for GRU-L2 and 0.616 ± 0.004 for the strongest GRU. Acc++ is a curriculum-dependent online diagnostic. On text8, L2C4 and L3C4 obtain 1.439 ± 0.003 and 1.435 ± 0.001 validation bits per character, versus 1.484 ± 0.007 for the strongest GRU and 1.452 ± 0.009 for a finite-context Transformer. An additional one-million-parameter series reproduces the improvement over GRU, but a local Mamba baseline performs better. These internal comparisons support a task-dependent benefit of modular recurrent organization, without isolating it from state capacity or establishing general superiority.

1 INTRODUCTION

Recurrent networks compress sequence history into a fixed-size state and update it incrementally. A conventional GRU (Cho et al., 2014) or LSTM (Hochreiter and Schmidhuber, 1997), however, represents each layer as a single hidden-state stream. Width or depth

adds capacity but leaves the organization and exchange of information within that state implicit.

How should a fixed recurrent parameter budget be organized into persistent modules with learned communication, and how do depth and column count affect outcomes? We distinguish memory organization, state capacity, and routing computation: components maintain local state while content-dependent routing moves information among them. A parameter-matched topology sweep can reveal non-monotonic relationships with state volume, while causal attribution requires additional state-matched controls.

We introduce *MoSAIC* (*Modular Self-Attentive Interacting Columns*), which distributes each layer across independently parameterized GRU columns. Each column queries a small message bank before its recurrent update. The bottom layer reads the current input and delayed top-layer states; higher layers read the newly updated states below. All columns update, and the top layer persists to the next timestep.

The term “Self-Attentive” refers to attention over a layer-specific bank of $C + 1$ messages at the bottom layer and C messages above, where C is the column count. This bank contains recurrent states and the current input; its size is independent of token history. Column and layer counts therefore organize a fixed-size recurrent state with dense synchronous updates.

We evaluate the original ten-million-parameter models on SDQ and text8, and add a one-million-parameter text8 series with recorded seeds, modern recurrent baselines, RIMs/BRIMs, and routing controls. Both series target one billion training tokens; comparisons use common logged budgets within each protocol. L2C4 is the compact two-layer reference, and L3C4 has the best observed MoSAIC mean. Short component ablations and exploratory LRU, associative-recall, and reinforcement-learning experiments test the boundaries of this result.

Our contributions are:

- **Architecture:** a layered grid of persistent recurrent columns with attention-mediated inter-column communication.
- **Topology result:** among the evaluated two-layer shapes at approximately fixed parameter scale, wider grids have more state and a higher SDQ diagnostic, but worse text8 BPC. Column count and within-column width change jointly.
- **Replication and boundaries:** improvements over GRU recur at two parameter scales; the smaller-scale series adds modular and modern baselines, sparse-routing controls, and component pilots. Mamba’s stronger text8 result and weak recall/RL transfer delimit the claim.

2 RELATED WORK

Modular recurrence and communication. Recurrent Independent Mechanisms (RIMs) (Goyal et al., 2021b) maintain separate recurrent modules, sparsely activate a subset for each input, and allow active modules to communicate through attention. BRIMs (Mittal et al., 2020) add hierarchical bottom-up and top-down communication, while shared-workspace models (Goyal et al., 2022) impose a communication bottleneck. Relational Memory Core (Santoro et al., 2018) updates interacting memory slots through attention. Object files and schemata (Goyal et al., 2021a) factorize persistent object states and shared update rules. Spatially Structured Recurrent Modules (Rahaman et al., 2021) learn interactions informed by spatial structure. MoSAIC shares bottom-up and delayed top-down information flow with BRIMs; its design combines independent GRUs, synchronous updates of every column, and a single-column readout. Our text8 experiments compare local RIMs/BRIMs implementations under a common recipe.

Grid and multidimensional recurrence. Multi-Dimensional RNNs (Graves et al., 2007) propagate information recurrently along multiple axes of structured data. Grid LSTM (Kalchbrenner et al., 2016) arranges LSTM blocks in a multidimensional grid whose dimensions exchange hidden and memory vectors through fixed connections. MoSAIC uses a grid as an organization of modules rather than as an additional axis of the input. All columns share one temporal axis, retain separate recurrent states, and communicate through content-dependent routing rather than edges fixed by input geometry.

Modern recurrent and attention models. Recent architectures redesign recurrent updates or memory representations: LRUs (Orvieto et al., 2023) use

stabilized linear recurrences, HGRN2 (Qin et al., 2024) expands gated linear state, Mamba (Gu and Dao, 2023) selects state-space dynamics from inputs, Mamba-2 (Dao and Gu, 2024) relates state-space models to structured attention, DeltaNet (Yang et al., 2024) uses delta-rule matrix state, and xLSTM (Beck et al., 2024) includes scalar and matrix memory. Transformers (Vaswani et al., 2017) and Transformer-XL (Dai et al., 2019) retain token representations for attention. Our new equal-token series compares local implementations of Mamba, HGRN2, DeltaNet, and mLSTM. The Mamba reference is a sequential selective-SSM implementation, not Mamba-2. Older reduced-token runs remain separate implementation context (Appendix A).

Routing and associative memory. Sparse mixture-of-experts layers (Shazeer et al., 2017; Fedus et al., 2021) select subsets of parameterized experts for individual inputs. MoSAIC routes messages among persistent columns that all update. Neural Turing Machines (Graves et al., 2014) learn attention-based access to external memory. Recall benchmarks (Arora et al., 2023) and induction-head analyses (Olsson et al., 2022) motivate associative retrieval tests. We distinguish the curriculum-dependent SDQ diagnostic from an additional multi-query associative recall (MQAR) pilot; neither establishes in-context learning in natural language.

3 METHOD

3.1 Persistent Columns and Layered Topology

Let $\mathbf{e}_t \in \mathbb{R}^{B \times H}$ be an embedded input for a batch of size B . A conventional L -layer GRU maintains $\mathbf{S}_t \in \mathbb{R}^{L \times B \times H}$. MoSAIC instead maintains

$$\mathbf{H}_t \in \mathbb{R}^{L \times C \times B \times H}, \quad (1)$$

where $\mathbf{h}_t^{l,c}$ denotes column c at layer l . Each layer-column pair has independent GRU parameters. Columns in a layer are evaluated together for efficiency, but do not share recurrent weights.

The message bank for the first layer contains the delayed top layer and the current external inputs:

$$\mathbf{M}_t^0 = [\mathbf{H}_{t-1}^{L-1}; \mathbf{e}_t] \in \mathbb{R}^{(C+1) \times B \times H}. \quad (2)$$

Each experiment uses one input stream. At subsequent layers,

$$\mathbf{M}_t^l = \mathbf{H}_t^{l-1}, \quad l > 0. \quad (3)$$

Information therefore moves bottom-up within a timestep, while the complete top layer provides a delayed feedback path across timesteps. Figure 1 summarizes this computation.

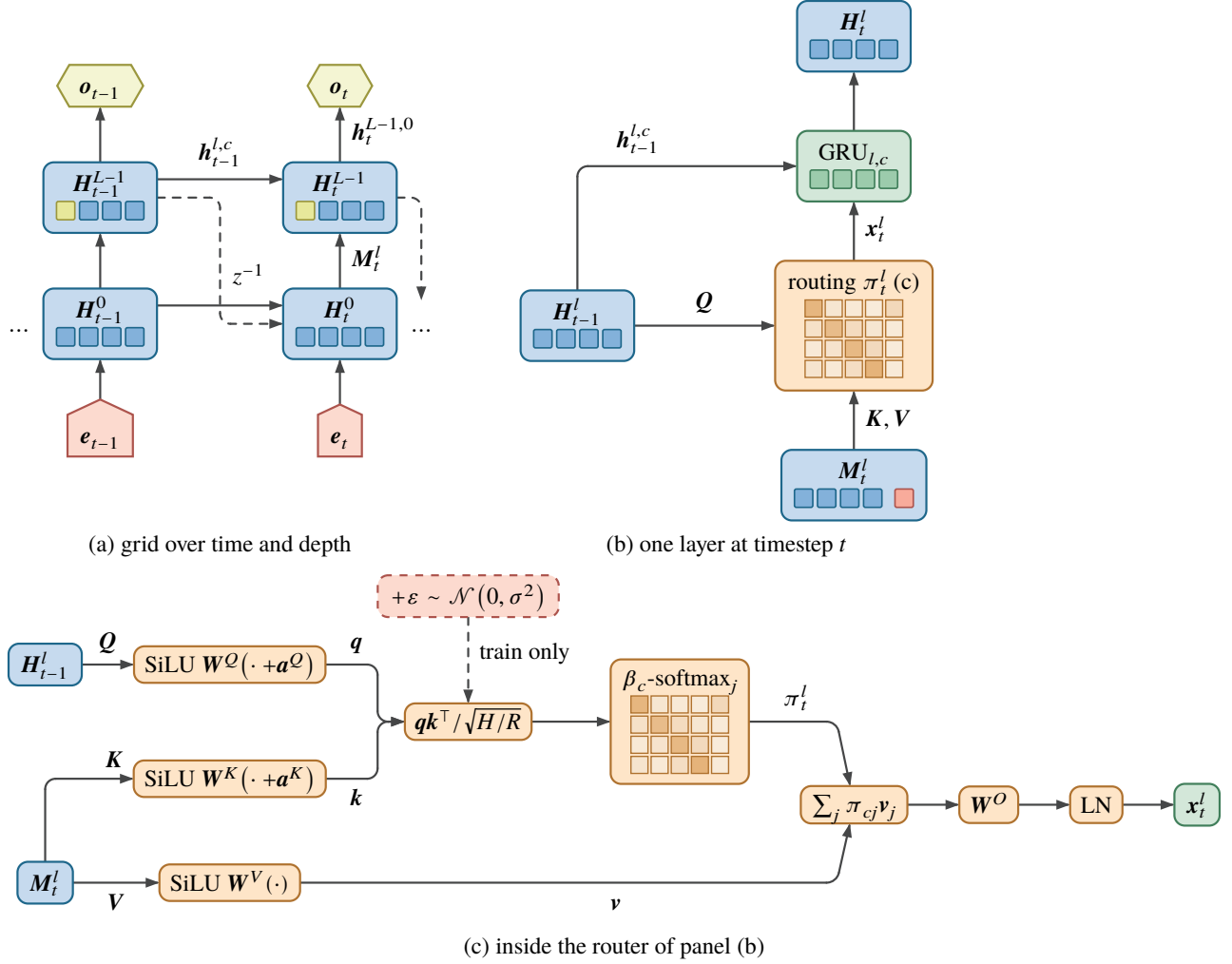


Figure 1: MoSAIC mechanism. At timestep t , each persistent column state queries a layer-specific message bank. The first layer reads the current input together with the previous top-layer column states; later layers read the newly updated states below them. The routed message is the input to an independent recurrent update. All top-layer column states are carried to timestep $t + 1$.

3.2 Attention-Mediated Routing

Layers and columns are indexed from zero. At layer l , the previous states of its C columns form the queries. Learnable identities $\mathbf{a}_c^{Q,l}, \mathbf{a}_j^{K,l} \in \mathbb{R}^H$ distinguish query columns and message sources. With R heads, each head has dimension $d = H/R$ and projection matrices $\mathbf{W}_r^{Q,l}, \mathbf{W}_r^{K,l}, \mathbf{W}_r^{V,l} \in \mathbb{R}^{d \times H}$ and biases in $\mathbb{R}^d$. The output projection has $\mathbf{W}^{O,l} \in \mathbb{R}^{H \times H}$ and $\mathbf{b}^{O,l} \in \mathbb{R}^H$. For head r ,

$$\mathbf{q}_{t,c,r}^l = \text{SiLU}(\mathbf{W}_r^{Q,l}(\mathbf{h}_{t-1}^{l,c} + \mathbf{a}_c^{Q,l}) + \mathbf{b}_r^{Q,l}), \quad (4)$$

$$\mathbf{k}_{t,j,r}^l = \text{SiLU}(\mathbf{W}_r^{K,l}(\mathbf{m}_{t,j}^l + \mathbf{a}_j^{K,l}) + \mathbf{b}_r^{K,l}), \quad (5)$$

$$\mathbf{v}_{t,j,r}^l = \text{SiLU}(\mathbf{W}_r^{V,l}\mathbf{m}_{t,j}^l + \mathbf{b}_r^{V,l}). \quad (6)$$

Each query column has a learned positive inverse temperature $\beta_c^l = \text{softplus}(\rho_c^l)$. The routing probabilities are

$$\pi_{t,c,j,r}^l = \text{softmax}_j \left[\beta_c^l \left(\frac{\mathbf{q}_{t,c,r}^l \cdot \mathbf{k}_{t,j,r}^l}{\sqrt{H/R}} + \epsilon_{t,c,j,r}^l \right) \right], \quad (7)$$

where R is the number of heads. The perturbation $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is independent during training and zero at evaluation. Noise is added before temperature scaling, so its effective logit standard deviation is $\beta_c^l \sigma_\epsilon$. SiLU (Elfwing et al., 2018) is applied to all three projections. Query and key identities distinguish participants; values carry projected content without an identity offset. These projection choices have not been individually ablated. Concatenated head outputs are lin-

early projected and layer-normalized:

$$\begin{aligned} \mathbf{z}_{t,c,r}^l &= \sum_j \pi_{t,c,j,r}^l \mathbf{v}_{t,j,r}^l, \\ \mathbf{u}_{t,c}^l &= \mathbf{W}^{O,l}[\mathbf{z}_{t,c,1}^l; \dots; \mathbf{z}_{t,c,R}^l] + \mathbf{b}^{O,l}, \\ \mathbf{x}_{t,c}^l &= \text{LN}(\mathbf{u}_{t,c}^l). \end{aligned} \quad (8)$$

Routing determines the input to each recurrent cell; it does not replace or post-process its persistent state.

3.3 Recurrent Update and Readout

Every column performs an ordinary independent GRU update,

$$\mathbf{h}_t^{l,c} = \text{GRU}_{l,c}(\mathbf{x}_{t,c}^l, \mathbf{h}_{t-1}^{l,c}). \quad (9)$$

The prediction head reads column 0 of the top layer. This intentional readout bottleneck requires useful information held in peripheral columns to be compressed and retrieved through one column. All top-layer columns persist and participate in the next bottom-layer message bank.

The full MoSAIC configuration uses three training-only routing terms: Gaussian logit noise, a communication cost favoring diagonal routes, and an entropy bonus favoring higher routing entropy. Let S_l be the number of messages at layer l . With zero-based column index c , the route cost $d_{c,j}^l$ is zero for the same-column recurrent message, one for another recurrent message, and $2c$ for the external input at the bottom layer. The communication loss is the mean expected route cost,

$$\mathcal{L}_{\text{comm}} = \mathbb{E}_{t,l,c,r} \left[\sum_{j=0}^{S_l-1} \pi_{t,c,j,r}^l d_{c,j}^l \right]. \quad (10)$$

Routing entropy is normalized by the log of the number of available messages,

$$\overline{\mathcal{H}}(\pi) = \mathbb{E}_{t,l,c,r} \left[\frac{-\sum_{j=0}^{S_l-1} \pi_{t,c,j,r}^l \log \pi_{t,c,j,r}^l}{\log S_l} \right]. \quad (11)$$

For task loss $\mathcal{L}_{\text{task}}$,

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{comm}} \mathcal{L}_{\text{comm}} - \lambda_{\text{ent}} \overline{\mathcal{H}}(\pi). \quad (12)$$

Full MoSAIC configurations use four routing heads. For text8, $(\lambda_{\text{comm}}, \lambda_{\text{ent}}, \sigma_\epsilon) = (0.05, 0.005, 0.05)$; for SDQ, the settings are $(0.02, 0.002, 0.02)$. Section 4.4 disables these terms individually and together in short text8 pilots; their contribution to the full-budget SDQ result remains untested. The LRU extensions use different routing and no auxiliary losses, so they are not isolated GRU-to-LRU cell substitutions.

3.4 Fixed-State and Resource Accounting

Attention is applied over $C + 1$ messages in the bottom layer and C messages above it. For fixed L , C , and H , the carried recurrent state and per-token routing geometry are independent of processed sequence length.

Ignoring embeddings and biases, the independent GRUs contribute approximately $6LCH^2$ parameters and routing projections approximately $4LH^2$. Persistent state contains LCH scalars per example, compared with LH for a stacked GRU. Parameter matching consequently does not match activation state or computation. Projection and GRU work is $O(L(C + 1)H^2)$ per example and token, while attention products require $O([C(C + 1) + (L - 1)C^2]H)$ work. Routing weights occupy $O(R[C(C + 1) + (L - 1)C^2])$ transient scalars per example, in addition to projections and GRU intermediates. Table 6 gives carried-state sizes and float32 storage; these are not peak training-memory measurements.

4 EXPERIMENTS

4.1 Protocols and Uncertainty

We separate two cohorts. The historical series compares approximately 10M-parameter models on SDQ and text8. The new series uses approximately 1M-parameter text8 models and recorded seeds; additional single-seed pilots investigate routing, linear cells, recall, and RL. A one-billion-token target is shared by the main text8 runs, but does not match compute or persistent state. All quality comparisons use validation or the explicitly identified online diagnostic.

Common-budget reporting. We retain the historical cohort fixed before this revision and reaggregate its raw curves. Historical text8 is evaluated at 960,036,864 logged tokens: the latest common recurrent validation step below every retained endpoint, including Transformer-64. Recurrent values are exact; Transformer values use linear interpolation between bracketing measurements without extrapolation. SDQ uses the same five logged points for every run, at 950,036,480 through 970,037,248 tokens. This replaces configuration-specific horizons and per-run final windows. Table 1 reports every historical main configuration.

The new text8 comparison uses exact shared steps without interpolation: 980,037,632 tokens in Table 2’s upper panel. Its lower panel explicitly labels separate horizons for Grid-LRU-L2 and Transformer; the depth-matched Grid-LRU-L2 comparison uses 960,036,864 tokens. Pilot protocols have their own

shared horizons. Logger steps identify retained training counters and can lag evaluation or the final flush; they establish comparable logged budgets, not identical wall time. We report mean $\pm$ sample standard deviation across launches, with denominator $n - 1$. Historical seeds were not recorded; new main rows use seeds 42/1337, adding 31337 where three runs are available. There are no significance tests, and one-seed pilots have no uncertainty estimate. Appendix E explains the limits of this reporting.

4.2 Historical Memory and Language Results

SDQ. Store-Distract-Query is an online associative-storage task with overwriting, distractors, and key-dependent counters. Its evolving curriculum increases episode length and decreases store/query frequency. We train with 512 streams, BPTT length 32, RMSprop, and gradient clipping at norm 1. Appendix B gives the generator and target exactly. Acc++ is accuracy on valid queries with a store-query gap strictly greater than the current batch’s mean valid gap. This is a changing subset of training-generator queries, not fixed-gap held-out accuracy.

At the common window, MoSAIC-L2C4 reaches 0.8389 ± 0.0062 , versus 0.5409 ± 0.0735 for GRU-L2 and 0.6159 ± 0.0040 for the strongest GRU, L1. L3C4 has the highest observed mean, 0.9137 ± 0.0123 . Every evaluated MoSAIC shape has a higher central diagnostic than every GRU depth. Figure 2(b) shows the trajectories; the changing curriculum limits interpretation as a fixed retrieval benchmark.

Text8. Text8 (Mahoney, 2011) contains 100M normalized English characters and a 27-character alphabet. Historical runs use the first 90M characters for training and the last 10M for validation, which was also used for manual configuration choices. Their BPC values are internal validation comparisons, not published text8 test results. Recurrent models use 512 contiguous streams, BPTT length 64, RMSprop, and the reset/learning-rate schedules in Appendix C. Evaluation disables random resets. BPC is cross-entropy in nats divided by $\log 2$.

At the common horizon, L2C4 obtains 1.4389 ± 0.0033 , versus 1.5005 ± 0.0121 for GRU-L2. L3C4 reaches 1.4349 ± 0.0008 , versus 1.4839 ± 0.0067 for the strongest GRU, L3. The cache-256 Transformer reaches 1.4522 ± 0.0088 . It shares the token batch, optimizer, and training target, but uses parallel segment processing and a rolling key/value cache. Its four layers have width 560, four heads, feed-forward width 1120, RMSNorm (Zhang and Sennrich, 2019), and RoPE (Su et al., 2021); cache lengths 256 and 64 are separate configurations. Figure 2(a) shows that the GRU-MoSAIC

separation is visible before the reporting horizon.

4.3 Modern and Modular References at Smaller Scale

The new series uses the contiguous 90M/5M/5M train/validation/test split. It retains 512 streams, 64-step BPTT, the common optimizer/reset recipe, and a 1B-token target. Models include embeddings and heads and range from 0.973M to 1.069M parameters (Table 2); matching is approximate, with state and compute separate. We evaluate the 5M validation segment with resets disabled; no test values were consulted to select architectures or construct these comparisons. A final test split is configured and some runs have logged test metrics, but we report validation throughout. The historical 10M validation segment includes this test segment, so the new split does not retroactively make it untouched by earlier development.

MoSAIC-L2C4 achieves 1.5966 ± 0.0050 BPC versus 1.6368 ± 0.0018 for GRU-L2; L3C4 reaches 1.5916 ± 0.0012 , versus 1.6205 ± 0.0076 for the strongest GRU, L3. The improvement therefore recurs at a smaller scale with recorded seeds. Local RIMs and BRIMs references obtain 1.8744 ± 0.0977 and 1.8968 ± 0.0318 . Each uses six LSTM modules, three active per token; BRIMs adds delayed top-down communication. They address the absence of direct modular references, but are compact reimplementations under our shared recipe, without architecture-specific tuning or a claim to reproduce their best published results.

Mamba is stronger than MoSAIC in this series, at 1.5651 ± 0.0012 BPC. HGRN2, DeltaNet, and mLSTM now have two runs at the same late token budget, replacing reduced-token comparisons at this scale. This is an internal comparison of local sequential implementations, not a ranking of optimized model families. The new cache-256 Transformer has 1.6396 ± 0.0098 BPC across two seeds at its own exact 982,646,784-token log; Table 2’s lower panel shows this separate horizon explicitly.

The simplified Grid-LRU reference replaces GRU cells, projected content attention, and auxiliary terms with complex diagonal linear recurrence and learned static mixing, retaining column-0 readout and normalized top-layer feedback (Appendix F). Thus it is a lighter architectural alternative, not a cell-only ablation. At a common 960,036,864 tokens, L2C4 Grid-LRU gives 1.6750 ± 0.0032 versus 1.6004 ± 0.0037 for MoSAIC-L2C4 ($n = 3$ each). L3C4 Grid-LRU has 1.6690 ± 0.0052 at the Table 2 horizon. Simpler updates do not preserve the text8 improvement; resource benefits require measurement.

Table 1: Historical $\sim 10\text{M}$ models, reaggregated at common logged budgets. Text8: 960,036,864 tokens; Transformer BPC is linearly interpolated between bracketing logs, recurrent rows are exact. SDQ: per-launch mean over the same five logs at 950.04–970.04M tokens. Values are mean $\pm$ sample SD; SDQ has three launches per row. Params are for text8. $\dagger$: two text8 launches. Acc++ is an online curriculum diagnostic.

Model	Params	Text8 n	Val. BPC $\downarrow$	SDQ Acc++ $\uparrow$
GRU-L1	10.16M	3	1.5641 ± 0.0027	0.6159 ± 0.0040
GRU-L2	10.04M	3	1.5005 ± 0.0121	0.5409 ± 0.0735
GRU-L3	10.23M	3	1.4839 ± 0.0067	0.3177 ± 0.0850
MoSAIC-L1C8	10.11M	3	1.4671 ± 0.0030	0.7009 ± 0.0186
MoSAIC-L2C4	10.11M	3	1.4389 ± 0.0033	0.8389 ± 0.0062
MoSAIC-L2C8	10.17M	3	1.4401 ± 0.0010	0.8641 ± 0.0129
MoSAIC-L2C16 $\dagger$	10.09M	2	1.4510 ± 0.0014	0.8830 ± 0.0116
MoSAIC-L3C4	9.99M	3	1.4349 ± 0.0008	0.9137 ± 0.0123
Transformer-256	10.07M	3	1.4522 ± 0.0088	—
Transformer-64	10.07M	3	1.4826 ± 0.0056	—

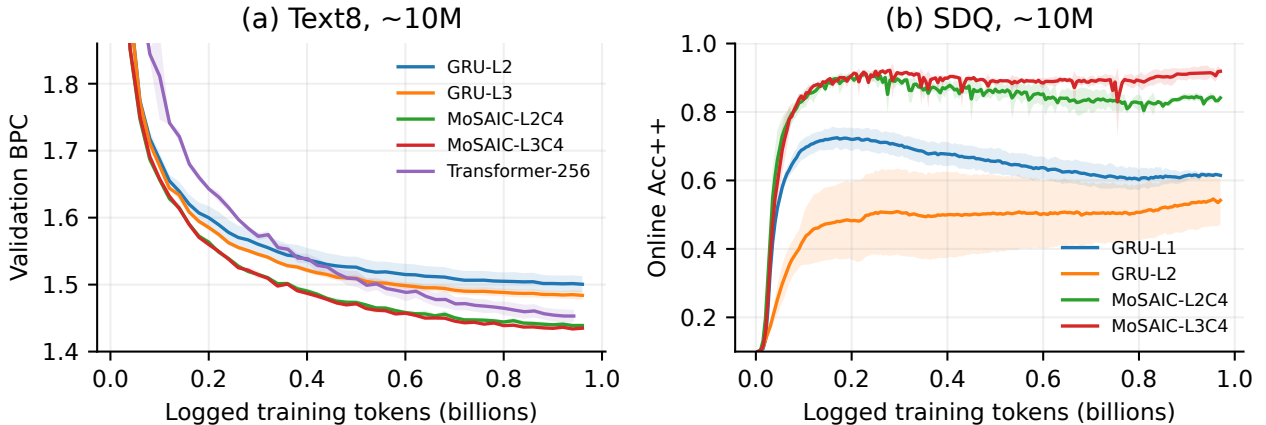


Figure 2: Historical $\sim 10\text{M}$ learning curves. Means and sample-SD bands across three launches on exact shared logs within each plotted configuration. Curves stop at the task’s common reporting horizon; Transformer measurements have their own logging grid. SDQ is an online curriculum diagnostic; historical text8 validation was consulted during configuration selection. Shading is descriptive, not a significance test.

4.4 Routing Sparsity and Component Pilots

Sparse message selection. Holding MoSAIC-L2C4 widths and routing terms fixed, seed-42 top- k attention retains only the k highest-scoring sources per query/head. At 980,037,632 tokens, dense routing gives 1.5921 BPC, top-2 gives 1.6353, and top-1 gives 1.8869. Restricting source access therefore hurts this text8 configuration, most strongly at top-1. All column cells still update: source sparsity is distinct from RIMs’ selective module updates and does not establish an execution speedup.

Training terms. Table 3 gives the first component test. Removing noise, communication cost, entropy bonus, or all three changes BPC by at most 0.0041 at this short common horizon, with signs changing across

earlier points. Removing the communication term nevertheless raises its logged expected route cost from about 0.896 to 1.045 at a shared 58.72M-token diagnostic, while normalized entropy stays near 0.96–0.97. The term changes routing without a resolved quality benefit in this pilot. The evaluation is near the end of the reset warmup, uses one seed and a capped validation prefix, and omits full-budget SDQ. Full-budget and repeated-seed ablations are still required to assess their quality contribution.

Longer BPTT and LRU mechanisms. Length-512 pilots keep 32,768 tokens/update by reducing training streams from 512 to 64. At 50.33M tokens they do not improve Grid-LRU, and GRU/MoSAIC are about 0.02 BPC worse; this does not isolate long-

Table 2: Text8 ~ 1 M series: mean $\pm$ sample SD of validation BPC. Upper panel: exactly 980,037,632 logged tokens. Lower panel uses separate horizons: $\dagger$ 960,036,864; $\ddagger$ 982,646,784, without interpolation. Seeds 42/1337, adding 31337 for $n = 3$. Split 90M/5M/5M. Local baselines share the training recipe; state and compute are unmatched.

Model	Params	n	BPC $\downarrow$
Mamba	1.003M	2	1.5651 ± 0.0012
MoSAIC L3C4	1.069M	3	1.5916 ± 0.0012
MoSAIC L2C4	1.050M	3	1.5966 ± 0.0050
MoSAIC L2C8	0.973M	2	1.6082 ± 0.0019
MoSAIC L1C8	1.034M	2	1.6369 ± 0.0016
GRU L1	1.004M	2	1.6807 ± 0.0003
GRU L2	1.014M	3	1.6368 ± 0.0018
GRU L3	0.986M	2	1.6205 ± 0.0076
Grid-LRU L1C8	1.004M	2	1.7473 ± 0.0005
Grid-LRU L2C8	1.060M	2	1.7156 ± 0.0047
Grid-LRU L3C4	1.063M	3	1.6690 ± 0.0052
HGRN2	0.996M	2	1.6695 ± 0.0005
mLSTM	0.996M	2	1.7163 ± 0.0039
DeltaNet	1.004M	2	1.7849 ± 0.0005
RIMs	0.995M	2	1.8744 ± 0.0977
BRIMs	1.000M	2	1.8968 ± 0.0318
Separate logged horizons:			
Grid-LRU L2C4 $\dagger$	1.049M	3	1.6750 ± 0.0032
Transformer-256 $\ddagger$	1.002M	2	1.6396 ± 0.0098

context capacity from trajectory diversity or training schedule (Appendix G). Five LRU memory/reading modifications and seven sparse-routing variants are reported with their own controls in Appendix F. Two hub reads improve text8 by 0.0157 BPC on one seed, but this gain does not transfer to our recall/RL pilots.

4.5 Additional Recall and RL Probes

MQAR (Arora et al., 2023) evaluates retrieval of multiple key-value pairs using an independent synthetic validation set. At a common 173.60M-token pilot horizon, the dynamic-hub LRU control answers 2.70% of scored queries overall: 24.825% with four stored pairs, falling to 0.142% with 64 pairs. All six variants have zero sequence-level exact match. The query-conditioned-read variant stays near chance; several runs have large gradient-norm spikes. Query-weighted aggregation emphasizes the largest-capacity bin, so raw training and validation accuracies are not directly comparable. These results expose limitations of the tested LRU memories; there are no completed strong-baseline MQAR comparisons (Appendix H).

In ten single-seed POPGym (Morad et al., 2023) pi-

Table 3: MoSAIC-L2C4 regularizer pilot, seed 42: validation BPC at 50,331,648 tokens. Same widths, optimizer and 524,288-token validation cap; Δ is relative to the full model. One launch per condition; no variability estimate.

Condition	BPC $\downarrow$	Δ
Full	1.9260	+0.0000
All three off	1.9282	+0.0022
No logit noise	1.9243	-0.0016
No communication cost	1.9220	-0.0040
No entropy bonus	1.9224	-0.0036

lots on RepeatPreviousHard and AutoencodeEasy, the Grid variants do not outperform plain LRU. On RepeatPreviousHard, the mean return across four common late evaluations is -0.3809 for LRU and about -0.49 to -0.50 for Grid models, near the random-policy expectation -0.5 . AutoencodeEasy results are also weak. These are exploratory transfer probes with a shared PPO recipe, not evidence of general RL superiority (Appendix I).

4.6 Topology, State, and Logged Costs

Parameter matching changes hidden width and carried state. Historical GRU-L2 stores 1,824 scalars, versus 3,392 for MoSAIC-L2C4; the new pair stores 576 versus 1,088. Tables 6 and 7 enumerate the counts. Within historical L2 MoSAIC, L2C16 stores 7,168 scalars but has higher text8 BPC than L2C4; SDQ improves with column count. This task-dependent non-monotonicity does not remove the GRU-versus-grid state confound. No completed state-matched quality comparison is available.

Resource logs cover 65 historical main and reduced-token launches on heterogeneous hardware. We restrict resource tables to the 48 confirmed H100 launches: Table 4 reports the main references and Appendix J gives the full H100 subset. Historical quality tables retain the original cohort. We reconstruct loop time as $\sum_i \Delta s_i / \text{perf} / \text{fps}_i$ through the last logged counter at or below 1B; effective rate is accounted tokens divided by time. These prefixes differ from the quality horizons. Time includes validation, compilation, inspection, and logging overhead; using one GPU type does not control contention. The measured context shows slower loop rates for these MoSAIC references than GRU/Transformer, without establishing controlled efficiency or exclusive GPU-hours.

Table 4: Historical log-derived costs for the H100 subset of the main references. Tokens, elapsed loop time, and effective rate are per-launch means $\pm$ sample standard deviations. †: one launch, with no variability estimate. This hardware subset has fewer launches than the quality tables; workload contention was not controlled. Time includes training-loop overhead and is not exclusive accelerator usage.

Task	Model	n	Logged tokens (M)	Loop time (h)	Rate (k tokens/s)
SDQ	MoSAIC-L2C4†	1	1000.0	1.327	209.3
SDQ	MoSAIC-L3C4	3	1000.0 ± 0.0	1.946 ± 0.311	145.4 ± 25.1
SDQ	GRU-L2†	1	1000.0	0.984	282.2
text8	MoSAIC-L2C4	2	1000.0 ± 0.0	1.627 ± 0.236	172.5 ± 25.1
text8	MoSAIC-L3C4	3	1000.0 ± 0.0	2.264 ± 0.033	122.7 ± 1.8
text8	GRU-L2	2	1000.0 ± 0.0	1.097 ± 0.016	253.1 ± 3.6
text8	Transformer-256†	1	1000.0	0.571	486.1

5 DISCUSSION AND LIMITATIONS

The evidence supports a specific result: under our approximately parameter-matched recipes, GRU-based modular state improves internal text8 validation at two scales and the historical SDQ training diagnostic. It does not establish an advantage over modern recurrent models: Mamba leads the new text8 comparison, simpler Grid-LRU loses quality, and the recall/RL extensions are weak. These comparisons match approximate parameter scale and logged tokens; state allocation and computation differ across architectures.

Column 0 is intentionally the sole readout. Its bottom-layer external-input route also has cost $2c = 0$, while column 15 pays 30 compared with cost 1 for an off-diagonal recurrent route. This asymmetry grows with column count. At fixed parameter budget, additional columns also reduce within-column and readout width (424 to 224 for historical L2C4 to L2C16). Routing cost and readout capacity therefore vary with topology; their individual contributions remain untested.

Historical seeds and the complete manual-tuning record are unavailable; historical text8 validation was reused during development, and SDQ lacks a fixed-gap held-out evaluation. The new series has recorded seeds and matched logged tokens, but two or three repeats are insufficient for broad uncertainty claims, and the baselines are local implementations under a shared recipe. Component and transfer pilots use one seed and cannot settle full-budget effects. State-matched and compute-matched controls, projection/readout ablations, and controlled latency/peak-memory measurements remain missing. Neither sparse message selection nor fixed-size state establishes faster execution, superior long-context scaling, or general Transformer replacement.

Availability. We will release code and experiment configurations publicly after acceptance. No anony-

mous code archive accompanies this draft. Training uses float32 with PyTorch compilation. New reported experiments used NVIDIA H100 80GB HBM3. Historical quality runs used heterogeneous hardware; their resource tables include only verified H100 launches (Appendix J).

6 CONCLUSION

MoSAIC organizes recurrence into persistent GRU columns that exchange messages through attention and expose a single-column readout. Common-budget historical comparisons and a smaller-scale replication support improvements over our GRU references, with task-dependent effects of state allocation. Modern baselines, sparse-routing controls, and negative LRU/recall/RL results sharpen the boundary of this finding; modular organization is useful in these protocols, while its causal mechanism and broader benefits remain open.

AI Use Statement

Generative AI tools were used to search, summarize, and analyze the research literature, and to assist in writing, debugging, and revising code. They also assisted architectural design, experiment configuration, analysis and interpretation of logged results, manuscript editing, bibliographic checks, figure-generation code, and conversion to the AISTATS LaTeX format. The authors retain responsibility for the implementation, analysis, references, and final manuscript content.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] Section 3 describes MoSAIC; Appendices F–I specify the extensions and pilot protocols.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes] Section 3 gives parameter, carried-state, and per-token routing complexity; no sample-complexity theorem is claimed.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [No] Code and configurations will be released after acceptance; no anonymous code archive is supplied with this draft.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Not Applicable] This paper presents no theoretical results.
 - (b) Complete proofs of all theoretical results. [Not Applicable] This paper presents no theoretical results.
 - (c) Clear explanations of any assumptions. [Not Applicable] Model assumptions are specified in Section 3; no theorem assumptions are required.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No] The public data source and protocol are described, but experiment code is not supplied with this draft.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [No] Sections 3 and 4 and Appendices C–I describe the retained configurations and new recorded seeds. The historical manual-tuning record and seed values are unavailable; local baselines have no exhaustive method-specific tuning.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes] Section 4, the captions, and Appendix E define common horizons, BPC, online Acc++, launch counts, and sample SD. One-seed pilots have no uncertainty estimate; their checkpoint averages are not independent training replicates.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] Sections 4.6 and 5 identify H100 for new runs and heterogeneous hardware for historical quality. Appendix J reports costs only for the 48 confirmed historical H100 launches, with actual subset counts.
 - (a) Citations of the creator If your work uses existing assets. [Yes] We cite text8, Zoology/MQAR, POPGym, and the original model and technique papers.
 - (b) The license information of the assets, if applicable. [No] A complete license inventory for the historical data and implementation dependencies has not been provided.
 - (c) New assets either in the supplemental material or as a URL, if applicable. [No] The implementation is planned for release after acceptance; no new asset is supplied with this draft.
 - (d) Information about consent from data providers/curators. [Not Applicable] We use an existing public text corpus and an algorithmic generator, without collecting new human data.
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [No] A separate sensitive-content audit of text8 was not conducted.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable] No crowdsourcing or human-subject study was conducted.
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable] No crowdsourcing or human-subject study was conducted.
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable] No crowdsourcing or human-subject study was conducted.

A REDUCED-TOKEN IMPLEMENTATION CONTEXT

HGRN2, DeltaNet, and mLSTM exceeded available memory at the standard 512-stream geometry and were run with smaller batches and training targets. Table 5 reports these historical runs separately. They are non-comparable reduced-token context (100–250M tokens versus the 1B target for the main rows) and are not used for any comparative quality or efficiency conclusion. Both tasks use two completed launches per reduced-token configuration. More optimizer updates do not compensate for differences in processed tokens, batch size, or the SDQ curriculum.

Table 5: Non-comparable reduced-token context. Updates are planned counts: target tokens divided by tokens/update. Final text8 BPC and final-five SDQ Acc++ are mean $\pm$ reported between-launch standard deviations ($n = 2$).

Task	Model	Target (M)	Tokens/update	Updates (k)	Metric
text8	HGRN2	100	2,048	48.8	1.6675 $\pm$ 0.0076
text8	DeltaNet	200	4,096	48.8	1.8280 $\pm$ 0.0231
text8	mLSTM	200	4,096	48.8	1.6797 $\pm$ 0.0084
SDQ	HGRN2	125	2,048	61.0	0.1095 $\pm$ 0.0004
SDQ	DeltaNet	250	4,096	61.0	0.1529 $\pm$ 0.0317
SDQ	mLSTM	250	4,096	61.0	0.1239 $\pm$ 0.0037

B SDQ GENERATOR AND HISTORICAL STATE INVENTORY

Task. SDQ is an online synthetic task for associative storage and retrieval under interference. With five keys and ten values, its vocabulary contains 50 key–value store tokens, ten distractor tokens, and five query tokens. Keys and values are sampled uniformly within the sampled token type. A store overwrites the value v_k for key k and increments its store count N_k^{store} . A distractor adds its value to a running sum D modulo ten. A query first increments N_k^{query} and then uses the target $y = (v_k + D + N_k^{\text{store}} - N_k^{\text{query}}) \bmod 10$. Stored values, counts, and D start at zero and reset at episode boundaries, together with recurrent state. Queries without a preceding store still receive a target but are excluded from Acc++. Cross-entropy is applied at every query and ignores store and distractor events.

Each stream resets independently before a token with probability $1/T$, giving geometric episode lengths. Every 200k processed tokens, the curriculum increases T by 0.125 up to 500, decreases p_{store} by 0.00014 down to 0.10, and decreases p_{query} by 0.00005 down to 0.25; initial values are $(T, p_{\text{store}}, p_{\text{query}}) = (10, 0.35, 0.35)$. Distractor probability is $1 - p_{\text{store}} - p_{\text{query}}$. Models use 512 streams, truncated backpropagation through time over 32 steps, RMSprop (Tieleman and Hinton, 2012), gradient clipping at norm 1, and the same learning-rate schedule, warming to 5×10^{-4} and decaying toward 5×10^{-5} . Appendix C specifies the schedules.

C TRAINING PROTOCOL DETAILS

The standard token batches are $512 \times 64 = 32,768$ for text8 and $512 \times 32 = 16,384$ for SDQ, giving approximately 30.5k and 61.0k planned optimizer updates respectively at the 1B target. Both tasks use float32, RMSprop with its PyTorch defaults ($\alpha = 0.99$, $\epsilon = 10^{-8}$, zero momentum and weight decay, uncentered), gradient clipping at norm 1, and no GRU dropout. Hidden widths, depths, column counts, and parameter counts are given in Table 6. Historical MoSAIC variants use four heads, independent GRUs, normalized routed messages, and the task-specific routing settings in Section 3. Historical seeds were null in the retained configurations, preventing exact seed replay or verification of seed independence.

The learning rate starts at 10^{-9} and warms linearly to 5×10^{-4} over 50 schedule events, one per 24 optimizer updates (1,200 updates). After warmup, every 250 optimizer updates it moves 1% of the remaining distance toward 5×10^{-5} . Text8 reset probability starts at 0.2/64 and every 100 optimizer updates moves 4% of the remaining distance toward 0.001/64. SDQ resets follow its generator’s evolving $1/T$ rather than this text8 reset schedule.

Table 6: Recurrent topology and carried-state storage per example (float32, 4 bytes/scalar; 1 KiB = 1024 bytes). Parameter counts are for text8; SDQ shares these widths and states. Storage excludes weights, optimizer state, intermediates, and caches.

Shape	H	State	KiB	Params
GRU-L1	1,296	1,296	5.06	10.16M
GRU-L2	912	1,824	7.12	10.04M
GRU-L3	752	2,256	8.81	10.23M
L1C8	440	3,520	13.75	10.11M
L2C4	424	3,392	13.25	10.11M
L2C8	312	4,992	19.50	10.17M
L2C16	224	7,168	28.00	10.09M
L3C4	344	4,128	16.12	9.99M

SDQ logs at approximately 5M-token intervals. Text8 validation is scheduled every 20M processed tokens and uses 512 contiguous streams with random resets disabled. For GRU/MoSAIC runs, it is capped at 20,000 sequential steps per stream, further limited by the validation stream length; states reset on a stream’s dataset wrap. The Transformer uses segment-based validation with its rolling cache. Validation BPC is mean cross-entropy in nats divided by $\log 2$. SDQ update-batch accuracies are logged through a bias-corrected exponential moving average with weight 0.005 per update (no reset at logging events), followed by the final-five summary. These are training-generator statistics, not an independent held-out set. The revised historical quality table uses common logged budgets as specified in Section 4; no held-out fixed-gap SDQ evaluation is available. The original configuration-specific summaries are retained in the local evidence record, rather than silently overwritten.

D NEW TEXT8 PROTOCOL AND BASELINE IMPLEMENTATIONS

The new main text8 runs use 512 streams, BPTT length 64, float32, RMSprop with the defaults specified above, and clipping at norm 1. Learning rate warms geometrically from 10^{-6} to 5×10^{-4} over 50 events spaced 12 optimizer updates apart, then every 250 updates moves 1% toward 5×10^{-5} . Reset probability warms geometrically from 0.1 to 0.2/64 over 100 events spaced 15 updates apart (49.152M tokens), then every 100 updates moves 4% toward 0.001/64. Validation is scheduled every 20M processed tokens with random resets disabled. The reported metric carries state within each validation stream. An additional reset-every-1024-token metric is logged separately and is not used in the main comparison. The common training recipe is shared; it is not a comprehensive architecture-specific optimizer search.

Table 7 gives counts for local implementations at the retained widths. Counts are analytical/accounting measurements, not peak memory. Mamba uses four layers, width 194, expansion 2, state dimension 16, convolution width 4, and a rank-one input-dependent time-step projection. This sequential implementation does not use the authors’ fused kernels. HGRN2 uses three layers, width 153, expansion 4; DeltaNet uses three width-217 layers; mLSTM uses three width-216 layers. The local matrix/gated-memory implementations and their normalization choices are shared-recipe references, not certified reproductions of official checkpoints. RIMs has one layer of six width-116 LSTM modules; BRIMs has two layers of six width-80 modules. Both activate three modules using attention to non-null input sources, preserve inactive recurrent states, and communicate through attention; BRIMs additionally uses delayed upper-layer modules. Transformer uses four width-176 layers, four heads, feed-forward width 352, and cache length 256. The separate two-seed Transformer mean is reported at its own logging grid in Section 4.3.

Figure 3(a) shows the new text8 trajectories and descriptive between-launch variability. Panel (b) separates MQAR retrieval by the number of stored pairs; the aggregate query accuracy weights these bins differently, as detailed in Appendix H.

E UNCERTAINTY AND EVIDENCE SELECTION

Means and sample standard deviations are calculated across replicate-level summaries, after applying the reporting horizon/window separately to each run. Standard-deviation bands in the figures describe between-launch variability and are not confidence intervals. With only two or three launches, neither normality nor a reliable

Table 7: New Text8 model accounting from the retained widths and local implementations, including token adapters ($V = 27$). State counts include necessary recurrent and public-message caches, or Transformer key/value caches; redundant views and bookkeeping are excluded. Float32 KiB exclude training intermediates and optimizer state.

Model	Width	Params	State scalars	KiB
Mamba	194	1,003,201	29,488	115.19
MoSAIC L3C4	112	1,068,743	1,344	5.25
MoSAIC L2C4	136	1,050,499	1,088	4.25
MoSAIC L2C8	96	972,715	1,536	6.00
MoSAIC L1C8	140	1,033,515	1,120	4.38
GRU L1	404	1,003,563	404	1.58
GRU L2	288	1,014,363	576	2.25
GRU L3	232	985,563	696	2.72
Grid-LRU L1C8	176	1,003,651	4,224	16.50
Grid-LRU L2C8	128	1,059,747	5,120	20.00
Grid-LRU L3C4	148	1,063,015	4,144	16.19
HGRN2	153	996,210	280,908	1097.30
mLSTM	216	995,577	140,619	549.29
DeltaNet	217	1,004,089	141,267	551.82
RIMs	696	994,859	1,392	5.44
BRIMs	480	1,000,395	1,920	7.50
Grid-LRU L2C4	180	1,049,463	3,600	14.06
Transformer-256	176	1,002,347	360,448	1408.00

population effect can be established. Historical seed independence and pairing cannot be verified from the retained records. New main runs have recorded seeds; the reported aggregate does not perform a paired significance test. Validation tuning, architecture selection, and multiple comparisons are not accounted for by the sample SD.

The historical cohort is fixed by the original retained run inventory; new main groups use the recorded-seed baseline inventory and frozen quality curves available for this revision. Horizons are chosen from common available logs, rather than from peaks or which step favors the proposed model. All historical shapes, new full-budget baseline groups, and the five LRU mechanism variants are reported. Pilots are explicitly separate from full-budget conclusions. Model/variant selection uses validation rather than test values. A tracker entry without a quality curve, or a metadata flag indicating that it is no longer running, is not treated as a completed quality experiment. In particular, short addressable-slot checks have no full-budget comparative evidence and are not included as recall results.

The common historical text8 horizon is 960,036,864 tokens. For the six Transformer launches, the bracketing measurements are 942,538,752 and 962,592,768 tokens; interpolation estimates BPC at the shared budget without pretending it is a measured checkpoint. All recurrent values there are direct observations. For SDQ, every launch contributes Acc++ at the same five counters, 950,036,480, 955,036,672, 960,036,864, 965,037,056, and 970,037,248. This resolves the earlier horizon/window mismatch, while leaving the changing Acc++ threshold and curriculum dependence unresolved.

F LRU EXTENSIONS AND SPARSE COMMUNICATION

Static Grid-LRU. The simplified baseline uses independent complex diagonal recurrences with residual public outputs. In schematic form, its private state obeys $\mathbf{s}_t = \boldsymbol{\lambda} \odot \mathbf{s}_{t-1} + \mathbf{B}\mathbf{x}_t$, where real and imaginary coordinates are packed in real tensors. Trainable pole radii are below one; initial e-folding horizons are sampled log-uniformly between 1 and 1,000 recurrent steps. The public output combines a projection of complex state with a direct residual input. Inter-column mixing is a learned row-softmax table independent of the current query, without the GRU model’s query/key/value projections, identities, stochastic routing, or auxiliary objectives. Bottom-layer feedback is parameter-free RMS-normalized. Only the top-layer column-0 public output reaches the prediction

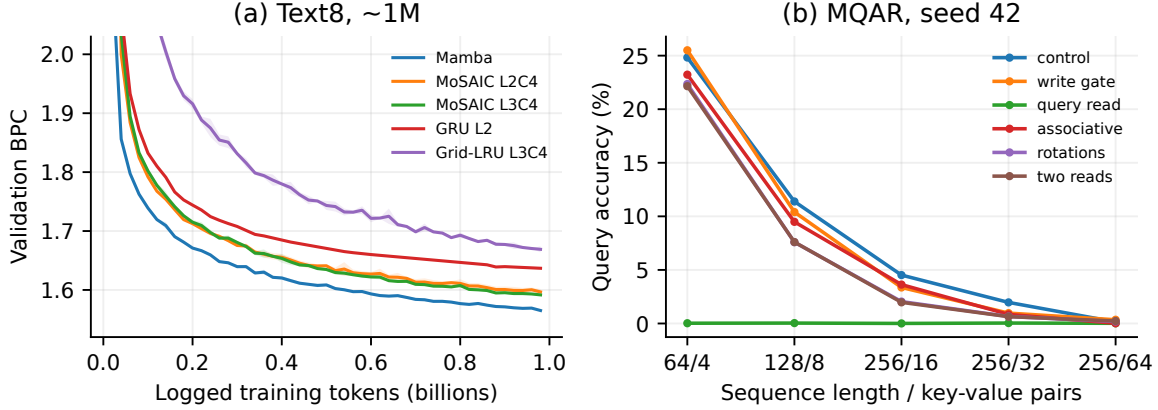


Figure 3: Additional evidence. (a) New text8 means and sample-SD bands on exact shared logs within each configuration ($n = 2$ for Mamba, $n = 3$ for the other plotted groups). (b) MQAR per-bin validation query accuracy for the six single-seed LRU pilots at their shared horizon. The text8 improvement from two reads does not imply reliable associative retrieval.

head. Stable poles constrain the isolated recurrence; they do not bound the coupled grid’s nonlinear feedback or guarantee stable training.

Five mechanism pilots. These use an L2C4H180 dynamic-hub control with star connectivity, an eight-dimensional query from the hub, zero-initialized query projection, normalized feedback, and no auxiliary loss. Peripheral routes remain query independent; the hub’s source weights depend on its previous public output. Every variant keeps the single-column prediction bottleneck. They therefore have a different control from the dense static Grid-LRU in Table 2.

- *Write gate*: a scalar sigmoid per peripheral column interpolates between its previous private state and the full candidate update, preserving real and imaginary memory when closed; the hub always updates. All projections still execute.
- *Query-conditioned read*: each peripheral private state is compressed to rank 16, multiplied by a hub-dependent gate, and projected back to the message width only for the hub’s read. This changes read contents in addition to source weights.
- *Associative column*: column 3 is replaced by a 16×32 matrix with normalized key $\mathbf{k}$, bounded value $\mathbf{v}$, and sigmoid write rate η . Its delta update is $\mathbf{M}' = \mathbf{M} + \eta \mathbf{k}(\mathbf{v} - \mathbf{M}^T \mathbf{k})^T$. A normalized hub query reads the matrix. This replacement changes both parameter count (about 0.830M versus 1.055M in text8) and state allocation, so it is not a matched-capacity control.
- *Rotations*: sequential Givens rotations exchange hub/peripheral real and imaginary coordinates before the recurrence, with four groups of angles initialized to zero. Norm preservation of this exchange alone does not establish grid stability.
- *Two reads*: after an ordinary all-column candidate pass, the hub reads freshly computed peripheral outputs and recomputes its candidate from the original previous hub state using shared weights. Peripherals write once. At four columns this adds approximately 25% cell-projection arithmetic, plus routing; it is not a measured wall-time estimate.

Two reads improves text8 from 1.6687 to 1.6530 BPC at the shared horizon; the paired late-window difference is also negative. Write gates and rotations change BPC much less, and query-conditioned/matrix reads worsen it. These are one-seed outcomes, not evidence of a universal memory improvement. The variants also differ in parameter counts, state allocation, or update computation, as specified above.

Table 8: Five LRU mechanisms and their own dynamic-hub control on Text8, seed 42, at 980,037,632 tokens. Late Δ averages paired differences at the common 800.03–980.04M logs. Reset-1024 evaluates the same validation data with state reset every 1,024 valid tokens; reset penalty is its BPC minus continuous BPC. Core widths are L2C4H180; added projections or replacement of an LRU by matrix memory change parameter/state counts. This is a single-seed exploratory comparison.

Variant	BPC ↓	Δ	Late Δ	Reset-1024	Reset penalty
control	1.6687	+0.0000	+0.0000	1.6803	+0.0116
write gate	1.6675	−0.0012	−0.0014	1.6799	+0.0124
query read	1.6872	+0.0185	+0.0184	1.7009	+0.0136
associative	1.6938	+0.0251	+0.0224	1.7121	+0.0183
rotations	1.6663	−0.0024	−0.0003	1.6787	+0.0123
two reads	1.6530	−0.0157	−0.0168	1.6641	+0.0111

Table 9: Sparse LRU pilots, seed 42. Each row uses its latest common logged point with the separate dense control; horizons differ across rows and are not cross-row rankings. Δ is paired BPC difference. Block has a 0.531M core versus approximately 1.05M for its dense control.

Variant	Tokens (M)	BPC ↓	Δ
self heavy	920.04	1.6725	−0.0063
star	1000.00	1.6816	+0.0055
star ring	600.02	1.7305	−0.0025
clockwork	920.04	1.7123	+0.0335
block	920.04	1.7607	+0.0819
entmax	1000.00	1.6861	+0.0099
dynamic hub	1000.00	1.6659	−0.0102

Periodic-reset diagnostic. The retained logs also evaluate the same validation data with each stream’s state reset every 1,024 valid tokens, independently of BPTT length. Table 8 reports this diagnostic at the same checkpoint. Two reads gives 1.6641 BPC versus 1.6803 for its dynamic-hub control, retaining a 0.0162-BPC advantage. The reset increases BPC by 0.0111 and 0.0116, respectively. This single-seed result shows that the gain persists under periodic resets; it does not establish superior retrieval beyond 1,024 tokens or identify the gain as a longer memory horizon. Late-window averages likewise summarize correlated checkpoints rather than independent repeats.

Sparse LRU pilots. The separate suite varies learned mixing initialization/connectivity: self-heavy mixing increases the initial diagonal preference; star routes peripherals through the hub; star-ring adds neighboring links; clockwork updates peripherals on scheduled slower clocks; block factorizes cell projections into smaller channel groups; entmax sparsifies routing probabilities; dynamic hub adds a query-conditioned hub read. All retain column-0 readout. These changes have different capacity and execution consequences, so Table 9 reports each with its own paired dense-control horizon and does not rank different-budget endpoints. The 1B final flush is recorded at 1,000,000,001 in the sparse suite.

Self-heavy and dynamic-hub pilots show small paired gains, while block and clockwork lose quality in the tested settings. The block core has about half the dense core’s parameters. Sparse attention support alone need not avoid dense projections or recurrent updates; none of these quality logs is a controlled throughput measurement.

G BPTT-LENGTH PILOTS

Table 10 compares ordinary GRU, plain LRU, MoSAIC-GRU, and static Grid-LRU at BPTT lengths 64 and 512 across eight launches. Each uses seed 42, the new text8 split, a target of 67,108,864 tokens, no test-based selection, and the same 524,288-token validation cap with 512 evaluation streams. Training streams change from 512 to 64 to keep the token batch at 32,768 and the planned update count equal. Reporting uses the shared 50,331,648-token validation checkpoint. The new text8 reset warmup ends near 49.152M tokens, so these pilots

Table 10: Rollout pilots at 50,331,648 tokens, seed 42; BPC with the same capped validation. Length 64 uses 512 training streams, length 512 uses 64: both have 32,768 tokens/update. Longer BPTT changes the number of independent streams and does not isolate context length alone.

Model	Rollout 64	Rollout 512	Δ
GRU	1.9826	2.0034	+0.0208
Grid-GRU	1.9260	1.9464	+0.0204
Grid-LRU	2.4867	2.4870	+0.0004
LRU	2.1575	2.1552	−0.0022

Table 11: MQAR pilots at 173,598,720 logged training tokens, seed 42: query-weighted validation accuracy (%) and cross-entropy (nats), followed by bin accuracies (%). All sequence-level exact-match accuracies are zero. Each bin contains 1,000 sequences; no strong-baseline or repeated-seed comparison is available.

Variant	Overall	CE	64/4	128/8	256/16	256/32	256/64
control	2.700	7.838	24.825	11.388	4.519	1.969	0.142
write gate	2.357	7.412	25.500	10.400	3.363	0.969	0.348
query read	0.023	8.322	0.025	0.037	0.006	0.041	0.017
associative	2.059	8.506	23.225	9.488	3.631	0.847	0.020
rotations	1.738	7.678	22.350	7.587	2.044	0.653	0.184
two reads	1.743	7.721	22.150	7.600	1.975	0.637	0.230

primarily describe early training, not convergence at long context.

GRU and MoSAIC-GRU are about 0.02 BPC worse at length 512; plain LRU changes by -0.0022 and Grid-LRU by $+0.0004$. This setting does not demonstrate a benefit from longer gradient propagation. A larger rollout also changes the number of independent trajectories per update and their resets; it cannot alone establish a memory-horizon effect.

H MULTI-QUERY ASSOCIATIVE RECALL PILOTS

The generator follows the single-pass MQAR protocol of Zoology (Arora et al., 2023). Vocabulary size is 8,192. Unique keys come from the lower vocabulary half and values from the upper half. A prefix stores alternating keys/values; each key is queried once at a distinct suffix position, with power-law position weighting exponent $0.01 - 1$. Other positions contain random vocabulary tokens. Only designated queries have supervised targets, at the position of the query key; filler coincidences are not scored. Training, validation, and test sample generators have separate fixed seeds (100/200/300); model/order seed is 42. Sequence state resets for each example, and BPTT spans the full sequence, so there is no truncation between stores and queries.

Training has 100,000 sequences at length/pairs 64/4 and 20,000 at each of 128/8, 256/16, 256/32, and 256/64. Validation has 1,000 independent sequences per training bin. The six pilots override the configured epoch count to eight, use Adam at 10^{-3} , batch size 32, norm clipping at 1, and query-normalized cross-entropy. At H180, the vocabulary adapters bring full model size to approximately 4M parameters; the core remains near the text8 mid scale, with the mechanism-specific differences noted above. This is not the same full-parameter budget as the text8 tables. We report the latest shared validation log at 173,598,720 tokens, rather than each model’s best checkpoint. Available logs do not establish completion of the full eight-epoch target or final test evaluation.

The overall accuracy weights each bin by its number of queries: with equal validation sequence counts, the five weights are 4/124, 8/124, 16/124, 32/124, and 64/124. Consequently the 64-pair bin contributes over half the aggregate, whereas the 4-pair bin dominates the training sequence count. For the control, the macro-average bin accuracy is 8.568%; reweighting validation by training sequence proportions gives about 15.79%, versus the logged training EMA near 21.12%. Neither is the reported query-weighted validation accuracy of 2.70%. This imbalance prevents interpreting their raw ratio as a generalization-gap estimate. Per-bin recall remains poor, including on training-range lengths.

Table 12: POPGym pilots, seed 42: mean evaluation return over the same four checkpoints at 6.291456, 6.815744, 7.340032, and 7.864320M vector slots. Each evaluation uses 32 stochastic-policy episodes with fresh environment seed 50000. Checkpoints share one training trajectory; their mean is not a seed-level uncertainty estimate.

Environment	Model	Late return $\uparrow$
RepeatPreviousHard	GRU	-0.5161
RepeatPreviousHard	LRU	-0.3809
RepeatPreviousHard	Grid-GRU	-0.4942
RepeatPreviousHard	Grid-LRU	-0.4973
RepeatPreviousHard	dynamic-hub	-0.4963
RepeatPreviousHard	two-reads	-0.4927
AutoencodeEasy	GRU	-0.4964
AutoencodeEasy	LRU	-0.4663
AutoencodeEasy	dynamic-hub	-0.5072
AutoencodeEasy	two-reads	-0.5111

Table 11 shows the per-bin retrieval results. Guessing uniformly over valid values gives $1/4096 \approx 0.0244\%$ accuracy; the query-read pilot is near this level overall. Its later loss reduction does not imply restored recall. Several runs log extremely large pre-clipping gradient-norm EMAs and some skipped updates; clipping and poles below one did not eliminate all numerical problems in these coupled recurrent models. These diagnostics are smoothed logger values, not exact per-update counts. Rotations avoids the large spikes in this cohort but has low retrieval accuracy. There are no completed repeated-seed GRU/Mamba/Transformer MQAR comparisons, so the pilot characterizes these LRU variants rather than ranking model families.

I POPGYM REINFORCEMENT-LEARNING PILOTS

We use POPGym (Morad et al., 2023) environments through the project’s RL runner: RepeatPreviousHard requires recalling a delayed past symbol, and AutoencodeEasy requires reconstructing an earlier observed sequence. This is a POPGym pilot, not evaluation on the separate MIKASA benchmark. Ten launches compare six models on RepeatPreviousHard (GRU, plain LRU, MoSAIC-GRU, static Grid-LRU, dynamic hub, two reads) and four on AutoencodeEasy (GRU, plain LRU, dynamic hub, two reads). Models have approximately 1M parameters and use seed 42; Grid readout remains restricted to column 0.

The common PPO setup has 32 environments, rollout length 128, 4,096 vector slots/update, two optimization epochs, $\gamma = 0.995$, GAE $\lambda = 0.99$, clipping ratio 0.1, value weight 0.5, entropy weight 0.005, gradient clipping at 0.5, and target KL 0.02. The learning rate warms from 10^{-6} to 2×10^{-4} over 16 events spaced four updates apart, then moves 1% toward 10^{-4} every 100 updates. The wrapper supplies previous action/reward identically to all models, normalizes readout features, and initializes policy-head gain to 0.01. Recurrent state is refreshed between optimization epochs; reset-only slots are masked. Budget 8,388,608 counts vector slots, not necessarily valid transitions. No communication auxiliary losses or recurrent logit noise are used.

Each evaluation uses a freshly created environment with seed 50000, stochastic actions, 32 completed episodes, and 16 evaluation environments. The four common reporting checkpoints end at 7,864,320 slots; they belong to one training trajectory and do not supply four independent training replicates. Evaluation uses a fixed seed across checkpoints and may repeatedly sample the same environment instances.

Table 12 reports the common-window returns. Random guessing has expected return -0.5 and perfect prediction $+1$ in these settings. Plain LRU’s RepeatPreviousHard mean -0.3809 is a modest departure from chance; it is not task mastery or proof of a particular memory strategy. Grid variants remain near chance on both probes. The ten runs have no logged skipped updates, but this does not establish convergence or robust stability. Some Grid-LRU raw states are much larger than those of plain LRU despite bounded poles and normalized feedback. These single-seed pilots require replication before drawing conclusions about RL performance.

J HISTORICAL RESOURCE ACCOUNTING

Resource logs were recovered for the 65 historical launches in the main and reduced-token quality summaries. The tables below restrict resource accounting to the 48 launches whose device logs identify NVIDIA H100 80GB HBM3; the other 17 launches used different GPU types and are excluded from these resource tables. Historical quality summaries retain the complete original cohort, including those launches. Thus hardware-restricted resource counts differ from quality-table counts. No device labels or measurements have been reassigned or converted to H100 equivalents.

For a run with processed-token counter s_i and interval rate $f_i = \text{perf}/\text{fps}_i$, we reconstruct $T_{\text{loop}} = \sum_i (s_i - s_{i-1})/f_i$, with $s_0 = 0$, through the last logged point at or below 1B. The effective rate is $s_{\text{last}}/T_{\text{loop}}$. The logger divides each token increment by elapsed time since its preceding flush. This includes compilation, training, validation, inspection, logging overhead between flushes, and any workload contention; it excludes finalization after the last flush. We do not extrapolate missing endpoints. These are log-derived elapsed-time proxies rather than dedicated throughput benchmarks or exclusive GPU-hours.

Update counts are $\lfloor s_{\text{last}}/(n_{\text{envs}} \ell_{\text{TBPTT}}) \rfloor$, counting reached update boundaries; a nonfinite-gradient skip need not perform an optimizer step. Tracker duration and the fps integral disagree in two launches in the complete historical inventory, so metadata duration is not used as the cost estimator. Tables report means $\pm$ sample standard deviations per launch; single-launch rows have no uncertainty estimate. H100 hardware labels do not control concurrent workloads or implementation differences. The new reported experiments also identify H100 80GB HBM3 in their device logs.

Table 13: SDQ: Main cohort resources, H100 subset only. Reduced-token rows have different batch and update geometries.

Model	n	Tokens (M)	Updates (k)	Loop time (h)	Rate (k tokens/s)
MoSAIC-L1C8	3	1000.0 $\pm$ 0.0	61.03 $\pm$ 0.00	1.126 $\pm$ 0.095	248.0 $\pm$ 22.1
MoSAIC-L2C16	3	1000.0 $\pm$ 0.0	61.03 $\pm$ 0.00	1.639 $\pm$ 0.144	170.4 $\pm$ 15.7
MoSAIC-L2C4	1	1000.0	61.03	1.327	209.3
MoSAIC-L2C8	3	1000.0 $\pm$ 0.0	61.03 $\pm$ 0.00	1.554 $\pm$ 0.138	179.7 $\pm$ 16.8
MoSAIC-L3C4	3	1000.0 $\pm$ 0.0	61.03 $\pm$ 0.00	1.946 $\pm$ 0.311	145.4 $\pm$ 25.1
GRU-L2	1	1000.0	61.03	0.984	282.2

Table 14: SDQ: Reduced-token context resources, H100 subset only. Reduced-token rows have different batch and update geometries.

Model	n	Tokens (M)	Updates (k)	Loop time (h)	Rate (k tokens/s)
DeltaNet	2	250.0 $\pm$ 0.0	61.03 $\pm$ 0.00	3.261 $\pm$ 0.042	21.3 $\pm$ 0.3
HGRN2	2	125.0 $\pm$ 0.0	61.03 $\pm$ 0.00	2.305 $\pm$ 0.017	15.1 $\pm$ 0.1
mLSTM	2	250.0 $\pm$ 0.0	61.03 $\pm$ 0.00	3.065 $\pm$ 0.026	22.7 $\pm$ 0.2

Table 15: text8: Main cohort resources, H100 subset only. Reduced-token rows have different batch and update geometries.

Model	n	Tokens (M)	Updates (k)	Loop time (h)	Rate (k tokens/s)
MoSAIC-L1C8	3	1000.0 ± 0.0	30.52 ± 0.00	1.213 ± 0.097	229.9 ± 17.6
MoSAIC-L2C16	2	1000.0 ± 0.0	30.52 ± 0.00	1.825 ± 0.123	152.6 ± 10.3
MoSAIC-L2C4	2	1000.0 ± 0.0	30.52 ± 0.00	1.627 ± 0.236	172.5 ± 25.1
MoSAIC-L2C8	3	1000.0 ± 0.0	30.52 ± 0.00	1.633 ± 0.146	171.0 ± 14.7
MoSAIC-L3C4	3	1000.0 ± 0.0	30.52 ± 0.00	2.264 ± 0.033	122.7 ± 1.8
GRU-L1	2	1000.0 ± 0.0	30.52 ± 0.00	0.925 ± 0.103	302.1 ± 33.6
GRU-L2	2	1000.0 ± 0.0	30.52 ± 0.00	1.097 ± 0.016	253.1 ± 3.6
GRU-L3	2	1000.0 ± 0.0	30.52 ± 0.00	1.240 ± 0.015	223.9 ± 2.6
Transformer-256	1	1000.0	30.52	0.571	486.1
Transformer-64	2	1000.0 ± 0.0	30.52 ± 0.00	0.365 ± 0.003	761.9 ± 6.4

Table 16: text8: Reduced-token context resources, H100 subset only. Reduced-token rows have different batch and update geometries.

Model	n	Tokens (M)	Updates (k)	Loop time (h)	Rate (k tokens/s)
DeltaNet	2	200.0 ± 0.0	48.83 ± 0.00	3.403 ± 0.011	16.3 ± 0.1
HGRN2	2	100.0 ± 0.0	48.83 ± 0.00	2.431 ± 0.006	11.4 ± 0.0
mLSTM	2	200.0 ± 0.0	48.83 ± 0.00	3.248 ± 0.027	17.1 ± 0.1